中南大学考试试卷

2009 -- 2010 学年 上 学期期末考试试题 A 卷 时间 100 分钟

科技英语 课程 32 学时 2 学分 考试形式： 开卷

专业年级： 2008 级选修课 总分 100 分，占总评成绩 70 %

注意：所有答案应按要求写在**答卷**上，做在试卷上的无效。

Part I Filling the following blanks with suitable words or phrases you have learned in the preceding units (10 Points)

1. **L**_____ an imaginary great circle on the surface of the earth passing through the north and south poles at right angles to the equator.
2. **L**_____, the angular distance between an imaginary line around a heavenly body parallel to its equator and the equator itself
3. **M** _____, the process of making something very small using modern technology
4. **E** _____, any of several complex proteins that are produced by cells and act as catalysts in specific biochemical.
5. **H** _____, An electrochemical cell in which the energy of a reaction between a fuel, such as liquid hydrogen, and an oxidant, such as liquid oxygen, is converted directly and continuously into electrical energy.
6. **S** _____, the result of converting sunlight into electricity.
7. **N** _____, the study of the control of matter on an atomic and molecular scale. Generally nanotechnology deals with structures of the size 100 nanometers or smaller, and involves developing materials or devices within that size.
8. **B** _____, a term referring to living and recently dead biological material that can be used as fuel.
9. **B** _____, an idea or plan produced by creative thought; invention
10. **L**_____, a personal computer designed for mobile use and small enough to sit on one's lap while in use.

Part II Based on the texts you have learnt, answer the following questions in English (20 Points)

1. What are the scientific success and technology success of European Space Agency's Smart-1 probe?
2. What do we mean by saying "the computer is disappearing"?
3. Why can we speed urban travel through removing roads and traffic lights?
4. What are the differences between traditional house and passive house?

Part III Translation (20 points)

1.The pervasiveness and near--invisibility of computing will be helped along by new technologies such as cheap, flexible displays, fingernail-sized MEMs (microelectromechanical systems) chips capable of storing terabytes of data.

2. If your conscience can withstand using a few drops of oil, the engine can heat and compress enough air to let the car get up to 90MPH.

3. Environmentalists everywhere have been salivating at the idea of running a car on vegetable oil ever since the idea was first proposed, and we have now taken the leap from concept to reality.

4. Ten to one（十之八九）, forms of energy come from the Sun directly or indirectly.

5. We know that a cat , whose eyes can take in many more rays of light than our eyes, can see clearly in the night.

Part IV Reading comprehension (20 Points)

Directions: There are two passages in this part. Each passage is followed by five questions. For each of them there are four choices marked [A], [B], [C]and [D]. You should choose the best answer.

Passage one

With the prospect of coal and petroleum supplies depleted and with air pollution becoming an increasing concern, the major countries of the world are seeking alternate sources of energy. If a means to obtain energy from water, especially from the ocean, can be effected economically, it would furnish a never ending supply of energy, since 70% of earth`s surface is ocean and another 10% is fresh water in rivers and lakes.

From the beginning of time man has used water power as a source of work-waterfalls and dams-but these are fresh water sources and are landlocked. The seas have contributed little or nothing in the way of power.

The use of temperature variation between currents is one area of exploration. Ocean water is heated by the sun near the equator and drawn by the rotation of the earth toward the poles, where it cools and drops toward the ocean floor and starts its journey back toward the equator. The differential between the two currents is 35 degrees to 45 degrees Fahrenheit and to use it the scientists must find the places where they run near land and are not too far away from each other. One area that meets these requirements is the Caribbean Sea.

Oceans also offer wave power, tides and the chemical propensities(倾向) of salt water as potential sources of energy. All these uses are theoretically possible. Britain `s Department of Energy is interested in wave power, using a string of “tear drop” devices that depend on very active wave areas and 100 feet depths. In additon to this , the British are working on a method that the Japanese have already put into pracitcal use on a small scale for powering their navigational buoys. This method is called an oscillating（震荡）

water column and rides the waves with a series of cylinders having one-way air valves. Wave movement produces air under pressure that has escape route –to a turbine(涡轮) that powers a generator.

1. The word "depleted"(Line 1, para. 1) possibly means "_____"

A. exhausted B discarded C polluted D disappeared

2. According to the first two paragraphs, _____.

A the use of water is a completely new concept

B the sea has long been used as a way of power

C water is a source of energy far more economical than any other energies

D water, if properly handled, may present a promising solution to energy crisis

3. According to the passage, the Caribbean Sea is the place where_____.

A two currents of different temperatures run near to each other and not far away from land

B two currents of different temperatures run near to each other but far away from land

C the ocean water is heated by the sun

D the ocean water gets cool

4. We can learn from the passage that _____.

A as far as energy source is concerned, salt water is useless because of its chemical properties

B the working principle of a vibrating water column is to use water as a direct power driving a generator

C Britain is pursuing a method that the Japanese have already put into practice.

D Britain is the most advanced country in developing new energy source.

5. What is the passage mainly about?

A Energy crisis.

B Water as a possible solution to energy crisis.

C History of the use of water as a source of work energy

D Properties of water

Passage Two

In managing information resources, the medium may be the key to an effective system. The medium is a vehicle, a tool, or a container for holding information; the information itself is the thing of value.

Three popular categories of information media are paper, film, and electronic storage devices. The media choice must not be viewed as a choice among these three, however, it must be viewed as an opportunity to select from a multitude of media possibilities in combinations that build effective systems. In many instances the person responsible for information-resource management is not the person who determines the medium in which information will be created. In such a case , the manager of a firm`s information resources faces a challenge in making a significant contribution to the organization`s objectives.

For effective management of information resources, media conversion may be necessary. Examples include keying or scanning paper documents to convert them to electronic media. Other processes convert electronic media from one format to another. For example, disk files created on one system may not be compatible with another system. Various hardware and software combinations can be used to convert files to formats that equipment will accept. For information generated within organizations, this necessicity of making systems compatible may be eliminated by cooperative planning. However, very little control can be exercised over the media used to generate information that comes to your organization from the outside.

The medium for information may be selected to satisfy a need that exists when information is created and communicated. For example, a paper record may be created because of its probability and because no special equipment is necessary; it is the fastest means of communicating information. A firm may use electronic mail because a network already exists for on-line computer communication. The additional application may cost less than postage to mail paper memos(备忘录)。

6. The first paragraph aims at telling the reader_____.

A the numerous resources of information

B the great variety of media for holding information

C the relationship between the medium and information

D the importance of information resources management

7. In the author 's opinion, which of the following statements is TRUE?

A People have three choices in selecting information medium.

B Paper is the best storage device.

C It is better to let the person responsible for information –resource management.

D The manager should build an effective system by selecting a good combination of different medium.

8. In order to achieve an effective management of information resources, the manager should_____.

A convert all paper documents to electronic media.

B make media conversion when necessary

C use one format in processing information.

D control the media used to generate information both inside and outside his organization.

9. Which of the following best expresses the main idea of the last paragraph?

A Different media for information may be selected for different purposes.

B By using different media, a firm can create various information for its objectives.

C Paper record is the most convenient medium for later reference.

D Electronic mail costs less than postage to mail paper memos.

10. Which of the following can best serve as the title of the passage?

A The Importance and Necessity of Media Conversion

B Three Categories of Information Media

C Various Means of Communication Information

D Media Selection in Managing Information Resources

Part V Writing (30 Points)

Please write the letter and the envelope to John Orton of Compax Computers in Hong Kong.

寄信人： 李坚强　　地址：成都电讯学院计算机系（邮编：610054）

收信人： John Orton　　地址：John Orton
Technical Consultant
Compax Computers
11, Fair Street
Hong Kong

尊敬的 Orton 先生：

您好！

今写信给您是有关贵公司上月送来的埃普生激光打印机(Epson Laser Printer)一事，并期望在互联网方面得到您的指导。

您一定记得 4 月 18 日我收到了送来的 25 台埃普生激光打印机。遗憾的是其中的 3 台有点问题，容易卡纸（the paper is not running through smoothly），且打印常常很不清楚。如果您能尽快派人来解决这些问题，我将不胜感激。

您能否向我们介绍一些现有的最新调制解调器。

早复为盼！

此致

敬礼！

李坚强

2000 年 5 月 19 日

大学英语四六级